Comparative Analysis of the Effects of PRM1 and PRM2 Gene Polymorphisms on Sperm Parameters in Men with Unexplained Infertility in South-West Nigeria

[*1,2]Ojerinde O. A., [2]Bakare O.C., [1]Agbolade O.M., [2,3]Onwuamah C.K., [1]Omidiran C.T., [1]Ojerinde O.V., [1]Aderemi E.O., [1]Ogidi V.N., [1]Sulaiman A.G., [1]George V.E., [1]Alagbe D.O., [4]Kemiki O.A.

[1]Department of Medical Laboratory Science, Federal University Oye-Ekiti, Ekiti State, Nigeria.
[2]Department of Biological Sciences, Lead City University, Ibadan, Oyo State, Nigeria.
[3]Nigerian Institute of Medical Research, Yaba, Lagos State, Nigeria.
[4]Babcock University, Ilishan-Remo, Ogun State, Nigeria.

ABSTRACT

Protamine 1 (PRM1) and Protamine 2 (PRM2) are essential for sperm chromatin condensation, and their disruption has been linked to impaired sperm function. Even though male factors account for nearly half of all infertility cases, comparative population-specific data on these genes are scarce. This study compared the effects of PRM1 and PRM2 polymorphisms on sperm parameters among men with unexplained infertility. A case–control study recruited 32 men each with unexplained infertility and fertile controls from South-West Nigerian. Semen was analysed and genomic DNA was extracted from blood, and the PRM1 (rs2301365) and PRM2 (rs1646023) polymorphisms were genotyped by real-time PCR allelic discrimination; 51 of 64 samples yielded successful genotypes (23 cases, 28 controls). Data were analysed in R (v4.5.1) using Mann–Whitney U test, Fisher's exact test, chi-square test, and Hardy–Weinberg equilibrium testing, with significance set at $p < .05$. Normal sperm morphology, motility, and volume were significantly lower among cases than controls while progressive motility and concentration did not differ significantly. The PRM1 polymorphism was in Hardy–Weinberg equilibrium ($\chi^2 = 0.2008$, $p = .654$) but showed no significant association with infertility at the genotype level (OR = 0.78, 95% CI: 0.17–3.48, $p = .749$) or allele level ($\chi^2 = 0.132$, $p = .716$). The PRM2 locus was monomorphic (100% CC genotype) in both groups, precluding association testing. Conclusively, men with unexplained infertility have measurably poorer sperm morphology, motility, and volume than fertile controls, but these differences could not be explained by PRM1 rs2301365 genotype.

Key words: Semen analysis, Protamine 1 and 2, Unexplained male infertility, Gene polymorphism, Chromatin condensation.

***Corresponding author.** Tel.+2348069423117: **E-mail address:** Kunle.ojerinde@fuoye.edu.ng.

1. INTRODUCTION

Infertility is estimated to affect one in six people of reproductive age globally, with male factors accounting for nearly half of all cases (1,2). Infertility is defined as the inability to conceive after at least twelve months of regular, unprotected intercourse. It carries significant emotional, social, and economic consequences in many African societies, yet the male contribution remains under-recognised despite evidence that men account for 40–50% of infertility cases (3). Male infertility remains under-researched in sub-Saharan Africa, where access to specialised diagnostic facilities is limited and evaluation is largely confined to routine semen analysis (4).

For decades, the diagnosis of male infertility has relied almost exclusively on conventional semen analysis of sperm motility, morphology, and concentration. These parameters, however, do not touch on the genetic competence of the spermatozoon (5). Many infertile men have normal semen parameters and thus classified as experiencing unexplained infertility and this

condition remains one of the most disturbing conditions in reproductive science, as the underlying defects are often molecular or genetic in nature and are not detected by routine testing (6).

Among the molecular contributors implicated in unexplained infertility, the protamine genes, Protamine 1 (PRM1) and Protamine 2 (PRM2), are of particular interest. During the final stage of spermatogenesis (spermiogenesis), the histones that package DNA in somatic cells are progressively replaced, first by transition proteins and then by protamines (7,8). This histone-to-protamine transition condenses the sperm genome into a near-crystalline structure that protects paternal DNA from oxidative damage and fragmentation as it transits the male and female reproductive tracts (9). The PRM1:PRM2 ratio is normally close to 1:1, and this balance is critical for proper chromatin condensation, sperm motility, and morphology (10).

Disruption of this balance, through dysfunctional gene expression, mutation, or polymorphism, has independently been linked to abnormal chromatin packaging, increased sperm DNA fragmentation, reduced motility, teratozoospermia, and diminished fertilising capacity (10,11). The interdependence of the two genes is further illustrated by murine evidence showing that loss of PRM1 alone produces deficient chromatin protamination, abnormal PRM2 processing, reduced sperm motility, and subfertility (11). Because PRM1 and PRM2 act on distinct but related chromatin-remodelling processes, each may plausibly affect individual sperm parameters; concentration, motility, morphology, DNA integrity, to differing degrees, a pattern that comparative, gene-level analysis is well placed to detect (12). Regulation of protamine expression by non-coding RNA adds a further layer of molecular complexity that may help explain variation in sperm quality among men who otherwise appear clinically similar (12).

Environmental and lifestyle factors compound these molecular vulnerabilities. Declining sperm quality and genomic instability at the population level have been associated with occupational exposure to endocrine-disrupting chemicals, oxidative stress, smoking, and advanced paternal age (13). Spermatogenesis is tightly regulated by the hypothalamic–pituitary–gonadal axis, and disruption of the cellular and molecular microenvironment of the seminiferous tubules can compromise the fidelity of chromatin remodelling during spermiogenesis (14,15). Such genetic abnormalities are thought to account for a meaningful proportion of male infertility, particularly among men with normal hormone levels, normal reproductive anatomy, and no history of procedures such as vasectomy (16).

Since advanced molecular tests, including gene expression profiling and PCR-based analysis, remain largely unavailable in the South West, couples are often treated without an identified genetic aetiology (4,17) and this diagnostic gap leaves a considerable number of unexplained male infertility cases unresolved, underscoring the need for region-specific molecular research to improve diagnosis and management.

2. MATERIALS AND METHODS

2.1. Study design: This study design is a cross sectional case–control design, involving men experiencing unexplained infertility as case and a control group of fertile men.

2.2. Study area: This study was conducted in the six states of South-West Nigeria namely; Lagos, Ogun, Oyo, Osun, Ondo, and Ekiti and sample analysis was done at the Nigerian Institute of Medical Research, Lagos.

2.3. Study population: Cases: Men aged 20–60 years experiencing unexplained infertility. **Controls:** Fertile Men aged 20–60 years.

2.4. Inclusion criteria: Men aged 20–60 years experiencing unexplained infertility as cases. Men aged 20–60 years with proven fertility as controls. Residents of South-West Nigeria willing to provide informed consent.

2.5. Exclusion criteria: Individuals unwilling to participate or provide informed consent. Men with a known cause of male-factor infertility.

2.6. Sampling technique and sample size: Participants were recruited using a purposive sampling technique. A total of 64 men were enrolled: 32 with unexplained infertility and 32 fertile controls. Sample size was calculated and validated using OpenEpi (version 3.01) at 80% statistical power.

2.7. Ethical considerations: Ethical approval for this study was obtained from the Institutional Review Board of the Nigerian Institute of Medical Research (NIMR; Project number: IRB/23/039). The study was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants following a full explanation of the study's aims, procedures, and their right to withdraw at any time without consequence, before specimen collection.

2.8. Sample collection: Semen samples were collected by masturbation into a sterile, wide-mouthed container, in accordance with WHO (2021) guidelines, and analysed immediately.

Five millilitres of venous blood was collected from each participant into an EDTA bottle, with each bottle assigned a unique identification number. Samples were centrifuged to separate plasma, which was then aliquoted into plain bottles and stored at -20°C until analysis.

2.9. Seminal fluid analysis: Liquefaction and macroscopic examination: Semen samples were allowed to liquefy at room temperature for approximately 30 minutes. Volume, colour, appearance, viscosity, liquefaction time, and pH were recorded, as these reflect the secretory function of the accessory glands.

Microscopic examination: A measured aliquot of liquefied semen was examined under the microscope to assess sperm concentration (spermatozoa per millilitre and total count), motility (proportion progressively motile, non-progressively motile, and immotile), morphology (proportion with normal head, midpiece, and tail structure by stained smear), vitality, and the presence of leucocytes or other cells indicative of infection or inflammation.

All recorded values were compared against the lower reference limits in the WHO (2021) laboratory manual, and each sample was classified as normal or abnormal accordingly.

2.10. Gene amplification and genotyping: Single nucleotide polymorphisms in the genes of interest were genotyped by real-time PCR allelic discrimination. Genomic DNA was extracted from the buffy coat of collected blood samples, and PCR amplification and genotyping were performed using gene-specific primers (18), probes, master mix, and Fast Optical 96-well reaction plates supplied by Thermo Fisher Scientific, USA.

Of the 64 participants' samples submitted for gene extraction, only 51 yielded genomic DNA of sufficient quality for genotyping. Of these 51 samples, the PRM1 locus (rs2301365) was successfully amplified and genotyped with evidence of allelic discrimination (G>T variants), whereas the PRM2 locus (rs1646023) yielded only the homozygous CC genotype, with no allelic variation detected across all 51 samples. As no variation was available for comparison, PRM2 was excluded from subsequent inferential analyses, though its genotype distribution is reported descriptively.

2.11. Statistical analysis: Data were analysed using R software (version 4.5.1). Descriptive statistics summarising baseline participant characteristics were computed and presented in frequency tables. The Mann–Whitney U test, a non-parametric test that compares two groups without assuming normality, was used for continuous variables. Associations between categorical variables were evaluated using the chi-square test or Fisher's exact test, as appropriate. Hardy–Weinberg equilibrium was assessed by a chi-square goodness-of-fit test.

Correlation analysis, logistic regression, and ROC analysis were performed where required. Statistical significance was set at $p < .05$ with 95% confidence intervals; all tests were two-tailed unless otherwise stated.

3. RESULTS

Characteristics of study participants: Table 1 shows no statistically significant differences between the control and case groups in age, educational level, religion, medical condition, or years in marriage (all $p > .05$). A statistically significant difference was observed in the number of children with a partner (median 3 vs 0; $p < .001$), consistent with the definition of the study groups.

Semen parameters of the case group relative to WHO lower reference limits: Table 2 compares the case group's semen parameters with the WHO (2021) lower reference limits. Virtually all values in the case group met or exceeded the WHO lower thresholds, supporting the classification of these participants as experiencing unexplained, rather than conventionally diagnosable, infertility.

Table 1: Demographic and background characteristics of study participants

Characteristic	Control (n=32)	Test (n=32)	p-value
Age group (years)			0.728[a]
20–30	1 (3.1%)	2 (6.3%)	
31–40	10 (31.3%)	11 (34.4%)	
41–50	15 (46.9%)	16 (50.0%)	
>51	6 (18.8%)	3 (9.4%)	
Education			0.394[a]
SSCE	4 (12.5%)	3 (9.4%)	
OND	2 (6.3%)	4 (12.5%)	
BSc	10 (31.3%)	15 (46.9%)	
MSc	14 (43.8%)	10 (31.3%)	
PhD	2 (6.3%)	0 (0.0%)	
Religion			0.750[a]
Christian	25 (78.1%)	27 (84.4%)	
Muslim	7 (21.9%)	5 (15.6%)	
Medical condition			1.000[a]
None	31 (96.9%)	31 (96.9%)	
Reported	1 (3.1%)	1 (3.1%)	
Years in marriage, Median (IQR)	10.0 (6.5)	8.0 (3.0)	0.241[b]
Children with partner, Median (IQR)	3.0 (1.0)	0.0 (0.0)	<0.001[b]

Percentages are within group. (a) Fisher's exact test with Monte Carlo simulation; (b) Mann–Whitney U test.

Table 2: Semen parameters of the case group compared with WHO lower reference limits

Parameter	Mean	SD	Median	WHO lower reference limit (95% CI)	Below limit, n (%)	Above limit, n (%)
Semen volume (mL)	2.68	1.32	2.50	≥ 1.5 (1.4–1.7)	1 (3.2)	30 (96.8)
Total motility (%)	59.38	7.80	60.00	≥ 40 (38–42)	1 (3.2)	30 (96.8)
Progressive motility (%)	43.44	7.01	45.00	≥ 32 (31–34)	2 (6.5)	29 (93.5)
Sperm concentration (×10⁶/mL)	52.77	42.60	31.00	≥ 15 (12–16)	1 (3.2)	30 (96.8)
Normal morphology (%)	25.48	4.15	25.00	≥ 4 (3–4)	0 (0.0)	31 (100.0)

WHO lower reference limits (95% CI) are from WHO (2021). Percentages are within the case group (n = 31).

Comparison of sperm parameters between groups: Table 3 and Figure 1 present the Mann–Whitney U comparison of five sperm parameters between the 32 fertile controls and 32 infertile cases. Normal sperm morphology showed the largest and most complete separation between groups (U = 992, p < .0001, r = -1.000), with means of 73.44% in controls versus 25.48% in cases. Total motility was significantly higher in controls (71.25%) than cases (59.38%; U = 770, p = .0001, r = -.552), as was semen volume (3.40 mL vs 2.68 mL; U = 656, p = .027, r = -.322). Progressive motility and sperm count did not differ significantly between groups (p = .147 and p = .096, respectively).

All fertile controls in this study had higher morphology, total motility, and semen volume than all infertile cases. This indicates that although the case group satisfied the absolute WHO minimum thresholds for these parameters, a distinct, statistically significant difference in sperm quality nonetheless existed between the two groups.

Table 3: Comparison of sperm parameters between control and case groups (Mann–Whitney U test)

Variable	Control (Mean ± SD)	Test (Mean ± SD)	U	P	R
Semen volume (mL)	3.40 ± 1.41	2.68 ± 1.32	656	.027	-.322
Total motility (%)	71.25 ± 13.56	59.38 ± 7.80	770	.0001	-.552
Progressive motility (%)	49.22 ± 13.51	43.44 ± 7.01	600	.147	-.209
Normal morphology (%)	73.44 ± 12.92	25.48 ± 4.15	992	<.0001	-1.000
Sperm count (×10⁶/mL)	69.34 ± 48.38	52.77 ± 42.60	618	.096	-.246

U = Mann–Whitney U test statistic; r = rank-biserial correlation (effect size). p-values are two-tailed.

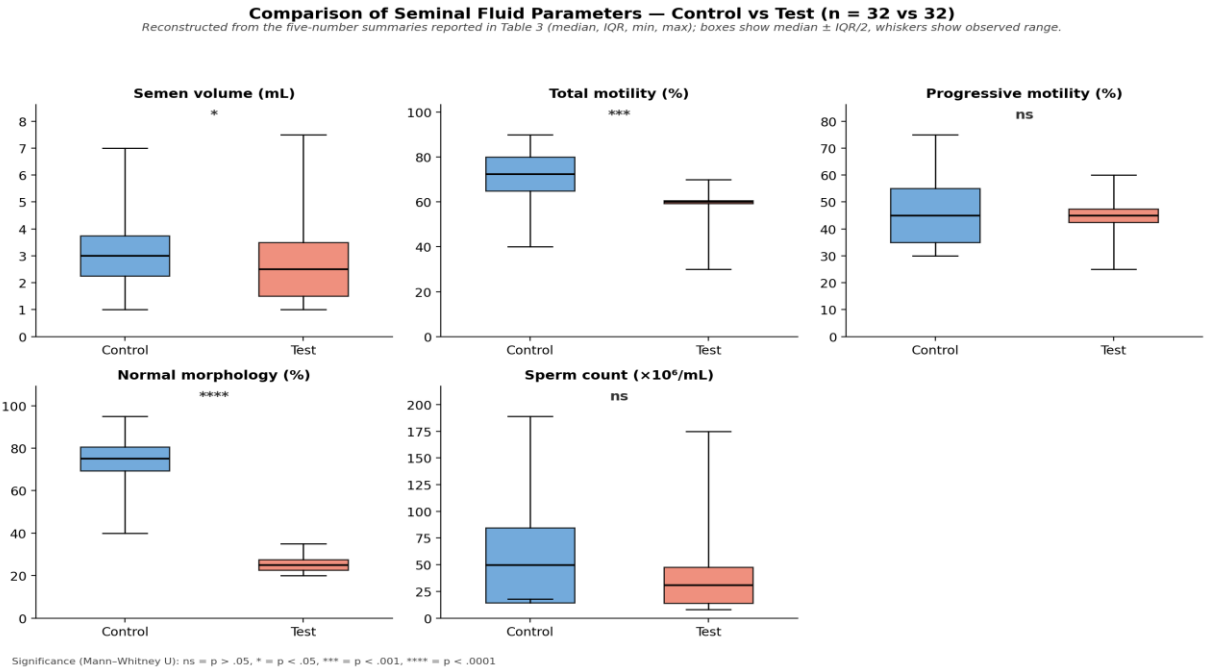


Figure 1: Comparison of sperm parameters between control and test groups.

Genetic polymorphisms of PRM1 and PRM2: Of the 64 participants enrolled, 51 were successfully genotyped for the PRM1 rs2301365 polymorphism (23 cases, 28 controls). Table 4 shows the genotype distribution, Table 5 the allele distribution, and Table 6 the Hardy–Weinberg equilibrium calculation.

The homozygous wild-type genotype (GG) was the most common, present in 39 of 51 participants (76.5%); the heterozygous genotype (GT) was present in 12 (23.5%). The homozygous variant (TT) was not detected in either group.

At the allele level (table 5), the G allele was dominant (frequency 0.8824; 90 of 102 alleles), with the T allele at 0.1176 (12 of 102 alleles).

The observed genotype frequencies did not differ significantly from those expected under Hardy–Weinberg equilibrium ($\chi^2 = 0.2008$, $p = .654$), indicating the sampled population was in genetic equilibrium at this locus (table 6).

All 51 genotyped participants (table 7), 23 in the case group and 28 in the control group, carried the homozygous CC genotype at the PRM2 rs1646023 locus, with no heterozygous (GC) or homozygous-variant (GG) genotype detected. Consequently, no allele or genotype frequency comparison was possible for PRM2.

Association between PRM1 and PRM2 polymorphisms and unexplained male infertility: To determine whether the PRM1 rs2301365 polymorphism was associated with infertility status, genotype- and allele-level analyses were performed (table 8 and 9). The genotype × group comparison (Fisher's exact test) yielded OR = 0.7767 (95% CI: 0.1725–3.4843, p = .749), and the allele × group comparison (chi-square test) yielded $\chi^2 = 0.1320$ (df = 1, p = .716; T-allele OR = 1.25, 95% CI: 0.374–4.173), neither statistically significant. The relative risk of the GT genotype in cases versus controls was 1.2174 (95% CI: 0.4532–3.2700), suggesting infertile men were approximately 22% more likely to carry the GT genotype than fertile men, though this excess risk was not statistically significant: the confidence interval spanned from a modestly protective effect to more than a threefold increase in risk.

Table 4: Genotype frequency distribution by group (PRM1 rs2301365)

Group	N	GG — n (%)	GT — n (%)	TT — n (%)
Case (Infertile)	23	17 (73.9%)	6 (26.1%)	0 (0.0%)
Control (Fertile)	28	22 (78.6%)	6 (21.4%)	0 (0.0%)
Total	51	39 (76.5%)	12 (23.5%)	0 (0.0%)

GG = homozygous wild-type; GT = heterozygous; TT = homozygous variant (not observed).

Table 5: Allele frequency distribution by group (PRM1 rs2301365)

Group	Total alleles	G allele — n (freq.)	T allele — n (freq.)
Case (Infertile)	46	40 (0.8696)	6 (0.1304)
Control (Fertile)	56	50 (0.8929)	6 (0.1071)
Total	102	90 (0.8824)	12 (0.1176)

Frequencies are allele proportions of total alleles per group; each diploid individual contributes two alleles.

Table 6: Hardy–Weinberg equilibrium test ($\chi^2 = 0.2008$, df = 1, p = 0.6541)

Genotype	Observed (n)	Observed freq.	Expected (n)	Expected freq.
GG	39	0.7647	39.71	0.7786
GT	12	0.2353	10.59	0.2076
TT	0	0.0000	0.71	0.0138

p(G) = 0.8824; p(T) = 0.1176. Chi-square computed on GG and GT classes (df = 1); TT expected n = 0.71.

Table 7: Genotype distribution by group (PRM2 rs1646023)

Group	N	CC — n (%)	GC — n (%)	GG — n (%)
Case (Infertile)	23	23 (100.0%)	0 (0.0%)	0 (0.0%)
Control (Fertile)	28	28 (100.0%)	0 (0.0%)	0 (0.0%)
Total	51	51 (100.0%)	0 (0.0%)	0 (0.0%)

No allelic variation was detected at this locus in either group.

Table 8: Genotype × group contingency table - Fisher's exact test

Genotype	Case (n=23)	Control (n=28)	Total (n=51)
GG (Wild-type)	17 (73.9%)	22 (78.6%)	39 (76.5%)
GT (Heterozygous)	6 (26.1%)	6 (21.4%)	12 (23.5%)
TT (Homozygous variant)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Fisher's exact test: OR = 0.7767, 95% CI 0.1725–3.4843, p = 0.7490. TT excluded (zero observations).

Table 9: Allele × group contingency table - chi-square test

Allele	Case — n (freq)	Control — n (freq)	Total — n (freq)
G (Wild-type)	40 (0.8696)	50 (0.8929)	90 (0.8824)
T (Variant)	6 (0.1304)	6 (0.1071)	12 (0.1176)

$\chi^2 = 0.1320$, df = 1, p = 0.7164. T-allele OR (Woolf method) = 1.25, 95% CI 0.374–4.173.

DISCUSSION

The first objective of this study was to confirm that gross semen abnormality was not responsible for the observed infertility. Relative to WHO lower reference limits, the case group's mean values for semen volume, total motility, progressive motility, and sperm concentration all exceeded the WHO minimum thresholds, with only one or two participants falling below each threshold; mean normal morphology (25.48%) likewise exceeded the WHO lower limit of 4%, indicating that none of the case-group men would be classified as teratozoospermic. This confirms that the case group met the diagnostic criteria for unexplained infertility, motivating a search for a molecular, gene-level explanation, since semen analysis alone did not account for their fertility difficulties (5).

The second objective was to compare sperm parameters between men with unexplained infertility and fertile men. Significant differences were found for three of the five parameters measured. Normal sperm morphology showed the largest difference between fertile controls (73.44%) and infertile cases (25.48%; $p < .0001$), consistent with its status as one of the semen parameters that most reliably distinguishes fertile from infertile men (1,19). Sperm with abnormal morphology, despite normal concentration, may be structurally unable to penetrate and fertilise an oocyte, which could partly account for the inability of case-group men to conceive despite an otherwise reasonable semen profile. Total motility was also significantly reduced in cases (59.38%) relative to controls (71.25%; $p = .0001$), as was semen volume (2.68 mL vs 3.40 mL; $p = .027$). Progressive motility and sperm count did not differ significantly, though both were numerically lower in cases, a pattern that may reflect natural inter-individual variation as well as the modest sample size available to detect smaller effects (20).

The third objective was to characterise the PRM1 and PRM2 polymorphisms in infertile and fertile men. For PRM1, the wild-type (GG) genotype was detected in 76.5% of participants and the heterozygous genotype (GT) in 23.5%, with no homozygous variant (TT) detected. This indicates that the T allele, while present, is uncommon in this population and occurs almost exclusively in heterozygous form, consistent with expectations for an essential chromatin-packaging gene (21,22). PRM1 genotype frequencies were in Hardy–Weinberg equilibrium ($\chi^2 = 0.2008$, $p = .654$), suggesting the genotyping assay was not subject to systematic error and that the sample was not markedly skewed at this locus (23).

All 51 participants genotyped for PRM2 carried the same homozygous CC genotype, with no variation in either group. This objective was therefore only partially achieved: genotyping was completed, but no distribution was available for comparison. This should not be read as evidence that PRM2 is unimportant to fertility, limited variation at this locus has been reported previously and linked to strong purifying selection during later protamine-processing steps (7,10,24). Because the widely reported PRM1:PRM2 ratio reflects gene expression or protein levels rather than DNA-level polymorphism, the absence of DNA variation at this locus does not imply that PRM2 expression was normal in these men (22,25).

The fourth objective was to examine the relationship between PRM1 and PRM2 polymorphisms and unexplained male infertility. Genotype analysis (Fisher's exact test) and allele analysis (chi-square test) were performed (26,27). For PRM1 rs2301365, neither the genotype analysis (OR = 0.7767, 95% CI: 0.1725–3.4843, p = .749) nor the allele analysis (χ^2 = 0.1320, p = .716; T-allele OR = 1.25, 95% CI: 0.374–4.173) showed a statistically significant association with infertility; all confidence intervals spanned the null value of 1. The relative risk for the GT genotype was likewise non-significant (RR = 1.2174, 95% CI: 0.4532–3.2700). Although the GT genotype and T allele were slightly more frequent among infertile men, this difference did not reach significance. This is broadly consistent with the established role of PRM1 in chromatin condensation and its reported association with elevated sperm DNA fragmentation (25,28); the small number of T alleles observed (12 in total) may have limited this study's power to detect a true but modest association (10). As all participants shared the same PRM2 genotype at the locus studied, no association test could be performed, and this part of the objective was only partially achieved. The absence of genetic variation at this locus should not be interpreted as evidence against a role for PRM2 in infertility, it indicates only that no difference was detectable at this particular SNP. A significant difference in sperm parameters existed between groups despite no significant association with PRM1, indicating that the semen-quality differences observed in infertile men cannot be explained by PRM1 variation alone. Male infertility is multifactorial, involving genetic, hormonal, environmental, and lifestyle contributors (29,30); PRM1 is therefore likely to be a contributing factor rather than the primary cause in this cohort.

5. CONCLUSION

The PRM1 rs2301365 polymorphism was in Hardy–Weinberg equilibrium in this South-West Nigerian population and was not significantly associated with unexplained male infertility at either the genotype or allele level. The monomorphism of the PRM2 rs1646023 locus in this population should be interpreted as an absence of detectable variation at this specific SNP, not as evidence against the well-established biological role of PRM2 in histone displacement and chromatin condensation, which is independent of this particular polymorphism.

Men with unexplained infertility in this cohort had significantly poorer sperm morphology, total motility, and semen volume than fertile controls, indicating measurable differences in semen quality between groups despite that most men with unexplained infertility had semen quality above the lower limits as stipulated by WHO and this supports the study's central finding that conventional semen analysis alone is an incomplete method for evaluating men with unexplained infertility.

Finally, the significant differences in sperm morphology and motility between groups could not be explained by PRM1 rs2301365 genotype, since genotype distribution did not differ significantly between cases and controls. This suggests that this specific polymorphism alone does not account for the poorer semen quality observed in the unexplained infertility group in this population.

Overall, the null hypothesis of no significant difference in PRM1 and PRM2 genetic status between men with unexplained infertility and fertile controls was accepted and the null hypothesis of no significant difference in sperm parameters between the two groups was equally accepted for morphology, motility, and semen volume. And the study achieves its central aim of generating original, locally derived molecular and seminal data on PRM1 and PRM2 in South-West Nigerian men.

Consent to participate: Written informed consent was obtained from all participants prior to enrolment.

Conflict of interest: The authors declare no conflict of interest.

Funding: This research received no specific grant from any funding agency.

REFERENCES

1. World Health Organisation. WHO laboratory manual for the examination and processing of human semen. 6th ed. Geneva: WHO; 2021. Available from: https://www.who.int/publications/i/item/9789240030787

2. Agarwal A, Baskaran S, Parekh N, Cho C, Henkel R, Vij S, Arafa M, Selvam MKP, Shah R. Male infertility. *Lancet,* 2021; 397(10271): 319–333.

3. Omolaoye TS, Dubon P, Omolaoye V, Du Plessis SS. Male infertility in Nigeria and South Africa: a ten-year observational study. *Andrologia,* 2023; 55(1): e14621.

4. Duvuru R, Halabi M, Omolaoye TS, Du Plessis SS. The genetic causes of male infertility: a Middle East and North Africa perspective. *F1000 Research,* 2022;11:125.

5. Esteves SC, Zini A, Coward RM, Evenson DP, Gosálvez J, Lewis SEM, Sharma R, Humaidan P. Sperm DNA fragmentation testing: summary evidence and clinical practice recommendations. *Andrologia,* 2020; 53 (2): e13874.

6. Leslie SW, Soon-Sutton TL, Khan MAB, Siref LE. Male infertility. In: StatPearls. Treasure Island (FL): *Stat Pearls Publishing,* 2022. Available from: https://www.ncbi.nlm.nih.gov/books/NBK562258/

7. Suede SH, Malik A, Sapra A. Histology, spermatogenesis. In: StatPearls. Treasure Island (FL): *StatPearls Publishing,* 2023.

8. Schlegel PN. Human spermatogenesis: insights from the clinical care of men with infertility. *Front Endocrinol.,* 2022; 13:889959.

9. Linn E, Ghanem L, Bhakta H, Greer C, Avella M. Genes regulating spermatogenesis and sperm function are associated with rare disorders. *Front Cell Dev Biol.,* 2021; 9:634536.

10. Nemati H, Sadeghi M, Nazeri M, Mohammadi M. Evaluation of the association between polymorphisms of PRM1 and PRM2 and the risk of male infertility: a systematic review, meta-analysis, and meta-regression. *Sci Rep.*, 2020; 10(1):17228.

11. Merges GE, Meier J, Schneider S, Kruse A, Fröbius AC, Kirfel G, Steger K, Arévalo L, Schorle H. Loss of Prm1 leads to defective chromatin protamination, impaired PRM2 processing, reduced sperm motility and subfertility in male mice. *Development,* 2022; 149(12): dev200330.

12. Peng L-F, Yu H. Circular RNA regulates male spermatogenesis: a narrative review. *Middle East Fertil Soc J.*, 2024; 29(1):12.

13. Skakkebæk NE, Lindahl-Jacobsen R, Levine H, Andersson A, Jørgensen N, Main KM, et al. Environmental factors in declining human fertility. *Nat Rev Endocrinol.*, 2021; 18(3): 139–157.

14. O'Donnell L, Stanton PG, de Kretser DM. Endocrinology of the male reproductive system and spermatogenesis. In: Endotext. South Dartmouth (MA): MDText.com; 2020.

15. Maroto M, Torvisco SN, García-Merino C, Fernández-González R, Pericuesta E. Mechanisms of hormonal, genetic, and temperature regulation of germ cell proliferation, differentiation, and death during spermatogenesis. *Biomolecules,* 2025; 15(4):500.

16. Lodish H, Berk A, Kaiser CA, Krieger M, Bretscher A, Ploegh H, Amon A, Scott MP. Molecular Cell Biology. 9th ed. New York: W. H. Freeman; 2021.

17. Şahin Y, Aslan ES, Aktuna S, Baltacı V. Evaluation of TNP1 and PRM1 gene expression in male infertility patients with low or high sperm DNA fragmentation. *J Turk Ger Gynecol Assoc.,* 2025; 26(1):7–14.

18. Niida Y, Togi S, Ura H. Streamlining genetic diagnosis with long-range polymerase chain reaction (PCR)-based next-generation sequencing for type I and type II collagenopathies. *Cureus,* 2023.

19. Menkveld R. Semen parameters, including WHO and strict criteria morphology, in a fertile and subfertile population: an effort

towards standardization of in-vivo thresholds. *Hum Reprod.*, 2001; 16(6):1165–1171.
20. Guzick DS, Overstreet JW, Factor-Litvak P, Brazil CK, Nakajima ST, Coutifaris C, et al. Sperm morphology, motility, and concentration in fertile and infertile men. *N Engl J Med.,* 2001; 345(19):1388–1393.
21. Oliva R. Protamines and male infertility. *Hum Reprod Update,* 2006; 12(4):417–435.
22. Rousseaux S, Caron C, Govin J, Lestrat C, Faure A, Khochbin S. Establishment of male-specific epigenetic information. *Gene,* 2005; 345(2):139–153.
23. Wigginton JE, Cutler DJ, Abecasis GR. A note on exact tests of Hardy–Weinberg equilibrium. *Am J Hum Genet.,* 2005; 76(5):887–893.
24. Tanaka H. Single nucleotide polymorphisms in the protamine-1 and -2 genes of fertile and infertile human male populations. *Mol Hum Reprod.,* 2003; 9(2):69–73.
25. Torregrosa N, Domínguez-Fandos D, Camejo MI, Shirley CR, Meistrich ML, Ballescà JL, Oliva R. Protamine 2 precursors, protamine 1/protamine 2 ratio, DNA integrity and other sperm parameters in infertile patients. *Hum Reprod.*, 2006; 21(8):2084–2089.
26. Fisher RA. On the interpretation of χ^2 from contingency tables, and the calculation of P. *J R Stat Soc.*, 1922; 85(1):87–94.
27. McDonald JH. Handbook of Biological Statistics. 3rd ed. Baltimore: Sparky House Publishing; 2014.
28. Aoki VW, Liu L, Carrell DT. Identification and evaluation of a novel sperm protamine abnormality in a population of infertile males. *Hum Reprod.,* 2005; 20(5):1298–1306.
29. Krausz C, Riera-Escamilla A. Genetics of male infertility. *Nat Rev Urol.,* 2018; 15(6):369–384.
30. Skakkebæk NE, Lindahl-Jacobsen R, Levine H, Andersson A, Jørgensen N, Main KM, et al. Environmental factors in declining human fertility. *Nat Rev Endocrinol.*, 2021; 18(3):139–157.